The two waves have different amplitudes a_1 and a_2, same angular frequency ω, and a phase difference of ϕ between them. The resultant displacement will be given by,

$$y = y_1 + y_2 = a_1 \sin \omega t + a_1 \sin (\omega t + \phi) \quad (7.8)$$

The simplification of the above equation by using trigonometric identities as done in (XI Physics 11.7) gives,

$$y = A \sin (\omega t + \theta) \quad (7.9)$$

$$\text{Where, } A = \sqrt{a_1^2 + a_2^2 + 2a_1 a_2 \cos \phi} \quad (7.10)$$

$$\theta = \tan^{-1} \frac{a_2 \sin \phi}{a_1 + a_2 \cos \phi} \quad (7.11)$$

The resultant amplitude is maximum,

$$A_{max} = \sqrt{(a_1 + a_2)^2}; \text{ when } \phi = 0, \pm 2\pi, \pm 4\pi \ldots, \quad (7.12)$$

The resultant amplitude is minimum,

$$A_{min} = \sqrt{(a_1 - a_2)^2}; \text{ when } \phi = \pm\pi, \pm 3\pi, \pm 5\pi \ldots, \quad (7.13)$$

The intensity of light is proportional to square of amplitude,

$$I \propto A^2 \quad (7.14)$$

Now, squaring equation (7.10) on both sides,

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi \quad (7.15)$$

In equation (7.15) if the phase difference, $\phi = 0, \pm 2\pi, \pm 4\pi \ldots$, it corresponds to the condition for maximum intensity of light called ***constructive interference***.

The resultant maximum intensity is,

$$I_{max} \propto (a_1 + a_2)^2$$
$$I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2} \quad (7.16)$$

In equation (7.15) if the phase difference, $\phi = \pm\pi, \pm 3\pi, \pm 5\pi \ldots$, it corresponds to the condition for minimum intensity of light called ***destructive interference***.

The resultant minimum intensity is,

$$I_{min} \propto (a_1 - a_2)^2$$
$$I_{min} = I_1 + I_2 - 2\sqrt{I_1 I_2} \quad (7.17)$$

As a special case, if $a_1 = a_2 = a$, then equation (7.10). becomes,

$$A = \sqrt{2a^2 + 2a^2 c \cos \phi} = \sqrt{2a^2 (1 + c \cos \phi)}$$
$$= \sqrt{2a^2 2\cos^2 (\phi / 2)}$$

$$A = 2a \cos(\phi / 2) \quad (7.18)$$

$$I \propto 4a^2 \cos^2 (\phi / 2) \quad [\because I \propto A^2] \quad (7.19)$$

$$I = 4\ I_0 \cos^2 (\phi / 2) \quad [\because I_0 \propto a^2] \quad (7.20)$$

$$I_{max} = 4I_0 \text{ when, } \phi = 0, \pm 2\pi, 4\pi \ldots, \quad (7.21)$$

$$I_{min} = 0 \text{ when, } \phi = \pm\pi, \pm 3\pi, \pm 5\pi \ldots, \quad (7.22)$$

We conclude that the phase difference ϕ, between the two waves decides the intensity of light at that point where the two waves meet.

EXAMPLE 7.2

Two light sources with amplitudes 5 units and 3 units respectively interfere with each other. Calculate the ratio of maximum and minimum intensities.

Solution

Amplitudes, $a_1 = 5$, $a_2 = 3$

Resultant amplitude,

$$A = \sqrt{a_1^2 + a_2^2 + 2a_1 a_2 \cos \phi}$$

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